

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 09

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>Correct answer only.</i> • B - radio waves do not travel through seawater; acoustic or satellite tags transmit position (1)	1
01.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved (2) • Radio / GPS tags or collars; leg rings / wing tags on birds; camera traps; DNA from droppings to confirm identity and diet (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
02.1	<i>1 mark for each point.</i> • k-selected: very slow growth, does not breed until 30, few eggs, so the population cannot replace losses (1) • Fish reach catchable size (20) before they breed (30) so fishing can remove the entire future breeding population (1) • Deep-water aggregations easily located and caught by trawling (1) • Recovery takes decades so fisheries have had to be closed (1)	4
02.2	<i>1 mark for each point.</i> • Herring breed at 2 years - the same age they become catchable - so there is always a breeding population and many young (1) • Cod are catchable at 2 but do not mature until 4, so heavy fishing removes most fish before they breed (1)	2
02.3	<i>1 mark for each point.</i> • Larger numbers of smaller / younger fish are caught, or bigger boats, nets and sonar increase catch effort (1) • Masks the decline of the breeding population until a sudden collapse (1)	2

Question	Marking guidance	Marks
02.4	<i>1 mark for each point.</i> - Neither is a target species but both are caught in demersal trawls; skate now extinct in the Baltic and critically endangered elsewhere (1) - Very late maturity (skate 11 years, Greenland shark 100-150 years, living over 400) and few young mean even low by-catch mortality exceeds replacement (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-342, 348-349; AQA spec 3.5.2.2.	
03.1	<i>1 mark for each point.</i> - Small clearings were farmed for a few years then abandoned as nutrients declined; the forest recolonised and soil fertility recovered before the area was cleared again (1) - Population growth shortens the gap between clearances so the forest and soil do not recover (1) - Clearings enlarge and degrade; less regrowth (1)	3
03.2	<i>1 mark for each point.</i> - Rainforest soils are thin and nutrient-poor (nutrients held in the biomass); once cleared, fertility falls in a few years and yields drop, so more land is cleared (1) - Abandoned land may not be recolonised if no forest survives nearby; erosion of exposed soil (1)	2
03.3	<i>1 mark for each point.</i> - Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing (1) - Reservoirs (eg Tucurui, 2850 km²) flood large areas, block migratory fish and river dolphins; decomposing vegetation produces acidic, deoxygenated water; fluctuating downstream flows (1) - Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees); cloud forests lose humidity as clouds form higher so frogs and epiphytes decline; some models predict replacement by savannah (1)	3
03.4	<i>1 mark for each point.</i> - A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt (1) - In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	
04.1	<i>1 mark for each point.</i> - Ecological footprint: the area of the Earth's surface (global hectares) needed to provide the resources used by a person / population / activity and to absorb the wastes produced (1) - Biocapacity: the biological productivity of an area (forests, cropland, grazing, fisheries, built land) - its ability to supply resources and absorb wastes, expressed in global hectares (1)	2
04.2	<i>1 mark for each point.</i> - A hectare of land / sea with the world-average biological productivity (1) - Biocapacities of real areas vary widely so a standard unit allows footprints and biocapacities of different places to be compared (1)	2

Question	Marking guidance	Marks
04.3	<i>1 mark for the correct answer.</i> • $8.1 / 1.7 = 4.8$ (accept 4.7-4.8) (1)	1
04.4	<i>1 mark for each point.</i> • Higher HDI is associated with a larger footprint (USA 8.1 vs Bangladesh 0.8), though the UK achieves the same HDI as the USA with about half the footprint (1) • Global footprint (2.8) exceeds global biocapacity (1.7) so humanity uses resources faster than they regenerate and CO₂ accumulates (overshoot) (1) • Earth Overshoot Day is the date each year when a year's worth of resources / waste absorption has been used up; it falls earlier each year (8 August in 2016) (1)	3
04.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Renewable energy and energy conservation (carbon footprint is the largest component) (1) • Reduced meat consumption / food waste; local seasonal food (1) • Circular economy: repair, re-use, recycling; longer product lives (1) • Public transport, efficient buildings (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 379-380; AQA spec 3.6.4.2.	
05.1	<i>Up to 4 marks in total; award marks in proportion to the completeness of each point.</i> • Rotation: each crop has its own pests; growing a different crop each year (4-5 year cycle) means pests left at the end of a season die before that crop returns, so populations cannot build up early (up to 2) • Beetle banks / hedgerows provide habitat and overwintering sites for natural predators (ground beetles, ladybirds) that eat aphids (up to 2) • Rotation also evens out nutrient demands and includes legumes (up to 2)	4
05.2	<i>1 mark for each point.</i> • Integrated control gives almost the same yield as chemical control (18.1 vs 18.5) with far fewer applications (1-2 vs 6) (1) • It retains most of the biodiversity of cultural control (26 vs 31 species) whereas chemical control alone leaves only 8 (1) • Cultural control alone loses about 12% yield; integrated control combines cultural methods, resistant varieties and carefully timed specific non-persistent pesticides only when essential - best balance of yield, cost and impact (1)	3
05.3	<i>1 mark for each point.</i> • Large numbers of males are sterilised with gamma radiation and released (1) • Females of many insects mate only once and store sperm, so mating with a sterile male produces no offspring; enough sterile males reduce fertile matings below replacement level (screw worm fly, Mediterranean fruit fly) (1) • Only the target species' females mate with them so non-target species are unaffected; fails if sterilised males behave abnormally or re-colonisation occurs (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 314-318; AQA spec 3.5.1.1.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> • Timber: building construction, furniture, tools, telegraph poles, concrete shuttering (oak, beech, mahogany, teak, conifer softwood) (1) • Fuel: fuelwood and charcoal - still the main energy source for most people in LEDCs (1) • Food: fruit, nuts, bushmeat, livestock fodder; crops of forest origin (coffee, cacao, bananas, Brazil nuts); pigs and chickens were forest animals (1) • Fibres: paper (wood pulp cellulose), cotton, viscose / rayon (1) • Medicines: quinine (cinchona), taxol (yew), aspirin (willow); most species not yet researched (1)	5
06.2	<i>1 mark for each point.</i> • Wood is mainly cellulose which is hard to digest, so carbon stored in wood has a long residence time; total forest carbon (vegetation plus soil) is about double that in the atmosphere (1) • Growing trees sequester CO₂ by photosynthesis and release oxygen; forest soils hold more carbon than the trees (especially boreal) (1) • Planting / protecting forests removes CO₂ and reduces the enhanced greenhouse effect (1)	3
06.3	<i>1 mark for each point.</i> • Highest transpiration of any ecosystem increases downwind rainfall; interception increases evaporation; deep organic soils retain water and delay runoff so river flow is more even and flooding reduced (1) • Low albedo absorbs sunlight; heat stored and released at night reduces day-night temperature extremes; canopy microclimate: lower light, wind, higher humidity (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-364; AQA spec 3.5.3.1.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Why needed: growing population (to 12 billion), land degradation since the fertile crescent, reliance on artificial fertilisers, pesticides and unsustainable irrigation; limits to new land and imports • Pest control: cultural methods (weeding, mulching, rotation, barrier / companion crops, predator habitats, biological control, sterile males, pheromones), resistant varieties, integrated control, reduced antibiotics • Nutrients: natural processes - nitrogen-fixing legumes, decomposition, recycling organic matter (manure, compost), crop rotation, permaculture, conservation of soil biota and mycorrhizae, reduced artificial fertiliser; phosphate recycling • Energy: natural processes instead of Haber-process fertiliser, low tillage, less machinery, renewable energy, low food miles, reduced methane (cattle diets, rice varieties) • Water: low water-use crops, drip irrigation, reservoirs and aquifer recharge, avoiding saline water; soil: organic matter maintenance, erosion control • Biodiversity: retaining hedgerows, ditches, ponds, woodlands; CWR conservation and seed banks; agri-environment schemes • Social: control of impacts on rural communities, Fairtrade, local food; consumer choices • Evaluation: yields vs sustainability, cost, need for policy support; judgement	9
07.2	<i>1 mark.</i> • Agricultural systems designed to mimic natural ecosystems - perennial crops, mixed species, recycling of nutrients, minimal external inputs (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 335-337; AQA spec 3.5.1.5.	

Question	Marking guidance	Marks
08.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
08.2	<i>1 mark.</i> • A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	
09.1	<i>1 mark for each point.</i> • The new opportunity is overwhelmingly attractive (economic benefit) (1) • Lack of knowledge / understanding of the impacts (1) • The impacts occur elsewhere or much later, so are not evident to those who benefit (1)	3
09.2	<i>1 mark for each point.</i> • Cultural belief that resources should be exploited to satisfy needs only (1) • Problems of over-exploitation become obvious quickly as local supplies dwindle, so behaviour changes (1)	2
09.3	<i>1 mark for each point.</i> • Resources are brought in from other areas (1) • If one area is depleted, exploitation moves to another area (eg timber, fish, metals, energy) without the consumers noticing (1)	2

Question	Marking guidance	Marks
09.4	<i>1 mark for each point.</i> • Easier access to energy has made many new activities possible (1) • The scale of activities has greatly increased (1) • Toxic wastes are produced that cannot be decontaminated by natural processes (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-371; AQA spec 3.6.1.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
10.2	<i>1 mark.</i> • The variety and abundance of life (species, genes, habitats) in an area / on Earth (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which agriculture can be made sustainable while feeding a growing human population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Need: population possibly 12 billion; humans raise carrying capacity; trophic level of diet • Impacts to be reduced: habitat loss, drainage, nutrient enrichment, introduced species, pesticide and nutrient pollution, nitrate toxicity, greenhouse gases, hydrological changes, soil erosion • Social, economic and political influences: consumer choices, technology availability, grants, guaranteed markets, surpluses, quotas, set-aside, agri-environment schemes • Strategies: cultural and biological pest control, integrated control, organic nutrient supply, legumes, low tillage, water conservation, CWR conservation, renewable energy, methane reduction, habitat retention • Limits: little new land, imports reducing local availability, cost and acceptance of new techniques, past land degradation • Justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-337; AQA spec 3.5.1.3-3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region • Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall) • Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling • Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study • Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	25